

Physiology: Pulmonary Circulation

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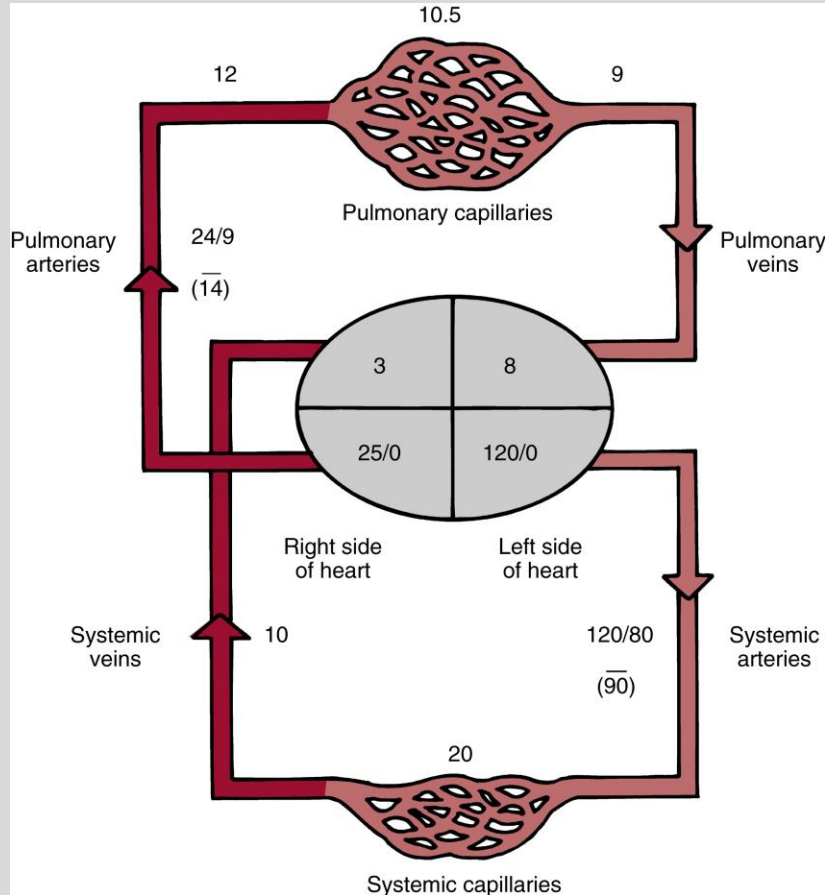


Do.
Make.
Innovate.
Reinvent the Future.

Objectives:

- Compare & contrast the functional anatomy of pulmonary vs. systemic circulation.
- Describe the pressures and flow in each system and their control points.
- Describe how pulmonary blood flow can be regulated. How is the autonomic nervous system involved?
- Recognize the factors controlling pulmonary vascular resistance and how they are expressed through Poiseuille's Law.
- Describe the effects of gravity on the distribution of pulmonary blood flow (Zones of West 1-3).
- Describe the interactions between alveolar, arterial, and venous pressures and the pulmonary alveolar pressure in determining flow in different regions of the lung.
- Compare pulmonary and extrapulmonary effects on pulmonary vascular resistance and blood flow.
- Describe the movement of solutes and gases across alveolar capillary barriers.
- Describe the pulmonary lymphatic system and how the Starling equation expresses lymphatic flow.
- Describe the active regulation of pulmonary blood flow by O₂ and other major vasoconstrictors and vasodilators.

Unique Dual Circulation



- Equal flow
- High vs low pressure
- 500ml in lungs
- 70ml in capillaries

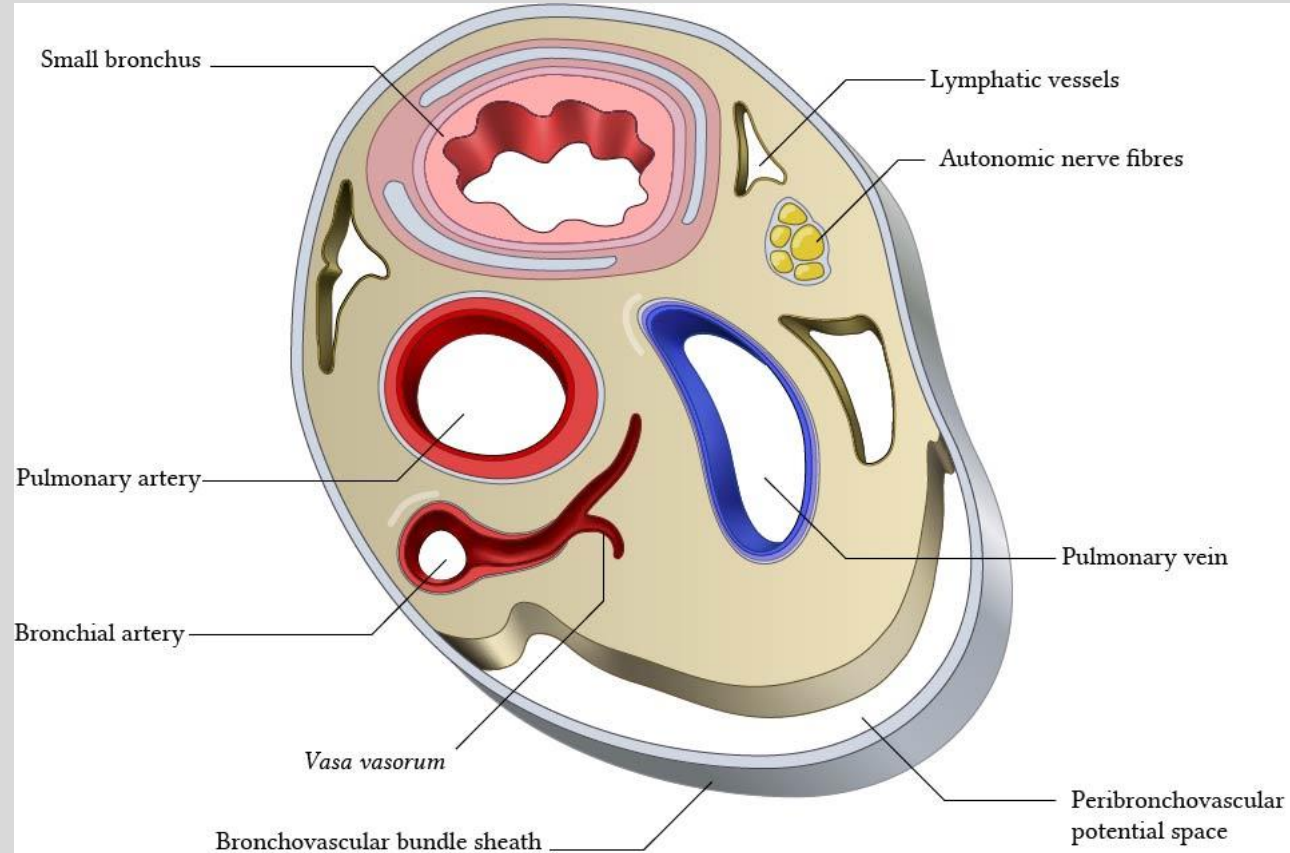
Reminder: **MAP= DP + 1/3(SP-DP)**
(there are other formulas as well)

DP = diastolic pressure

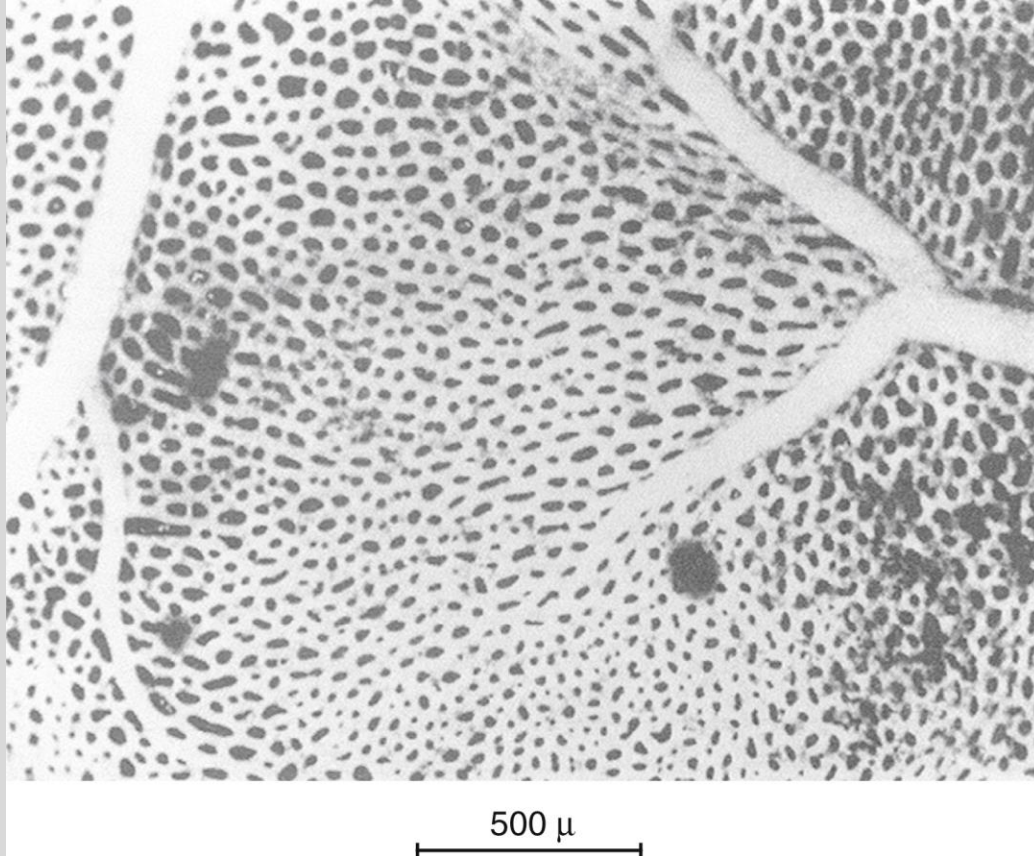
SP = systolic pressure

MAP = mean arterial pressure

Bronchovascular Bundle



Alveolar Capillaries - A Sheet of Blood

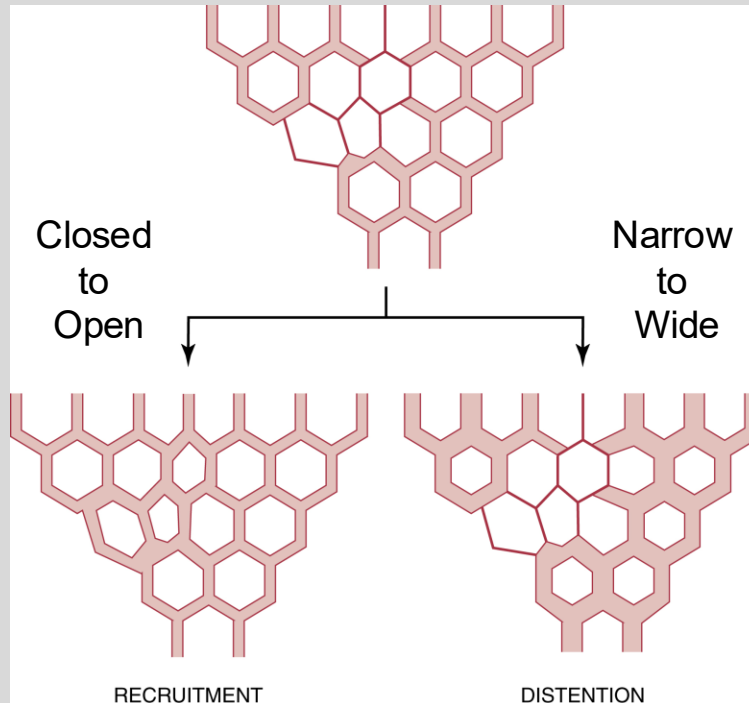


- 280 billion capillaries/300million alveoli
- Surface area for gas exchange 70 - 80m² (~800 sq feet)
~ 5 parking spaces
- 0.75 seconds to passage
- 6 μm diameter (RBC ~8 μm)

Blood Flow Adjustments

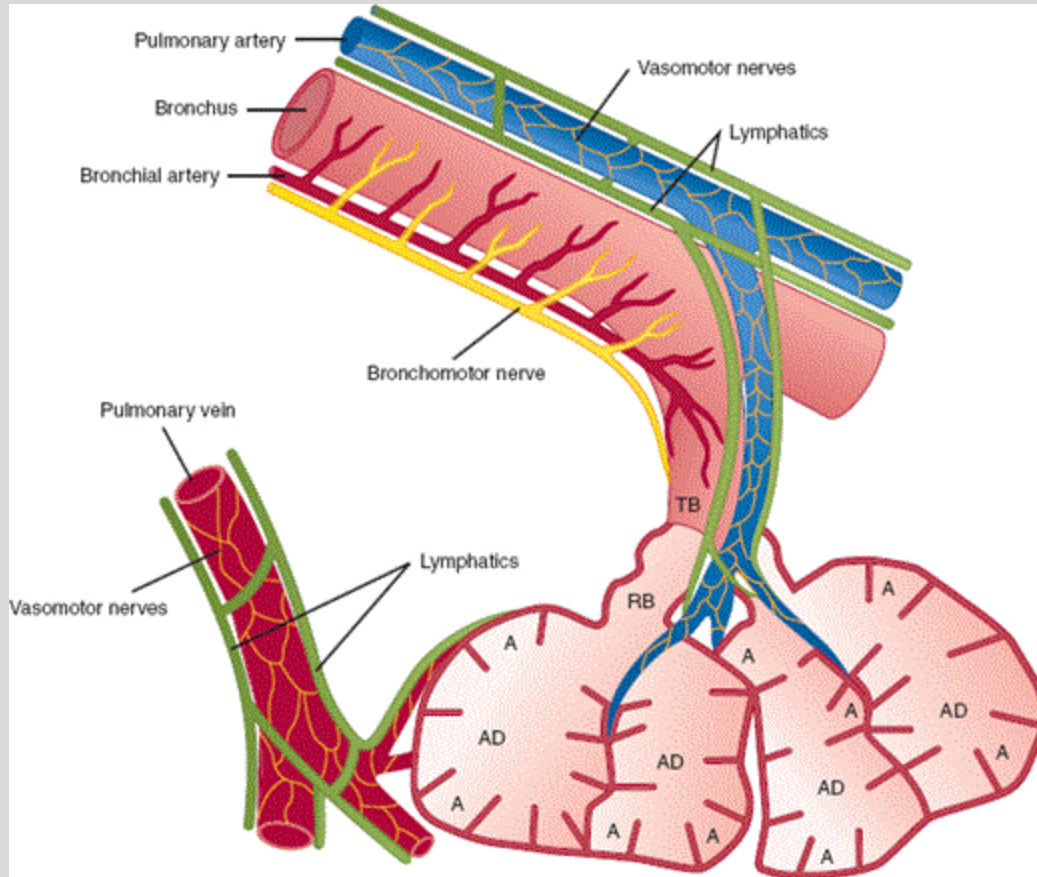
3.5 L/min/m² (at rest) → **25 L/min/m²** of body surface area during exercise

70 mL (at rest) → **200 mL** in pulmonary capillary bed



No significant effect on pulmonary resistance!

Sympathetic Regulation of Pulmonary Blood Flow

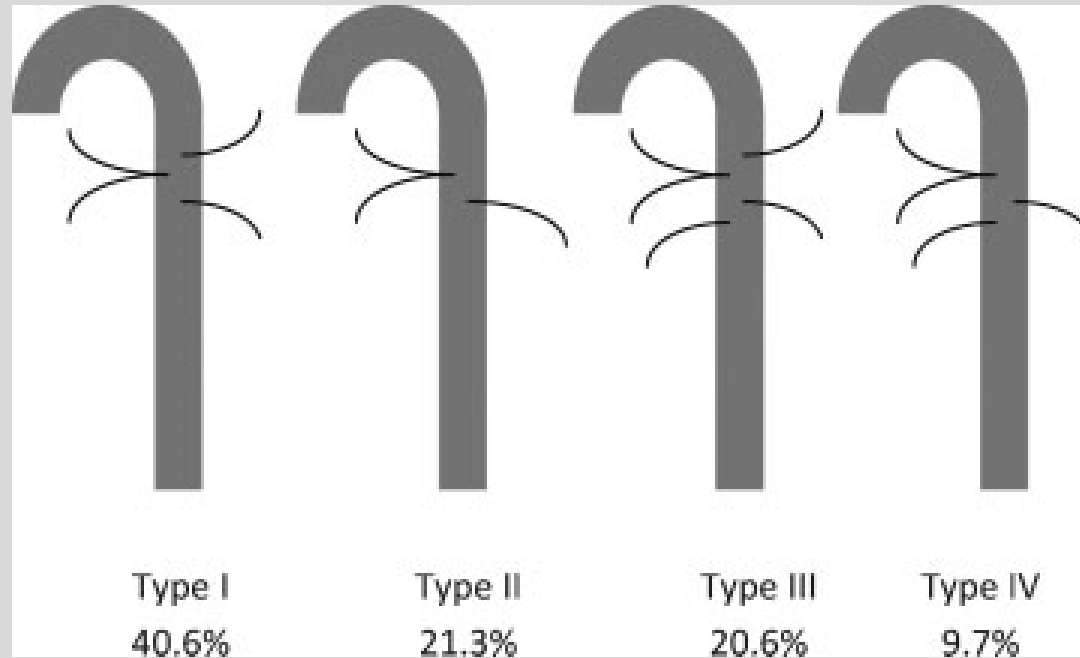


Pulmonary **arteries & veins** $> 50 \mu\text{m}$
have smooth muscle

Pulmonary arterioles and capillaries
have no smooth muscle

Bronchial Artery Anatomy

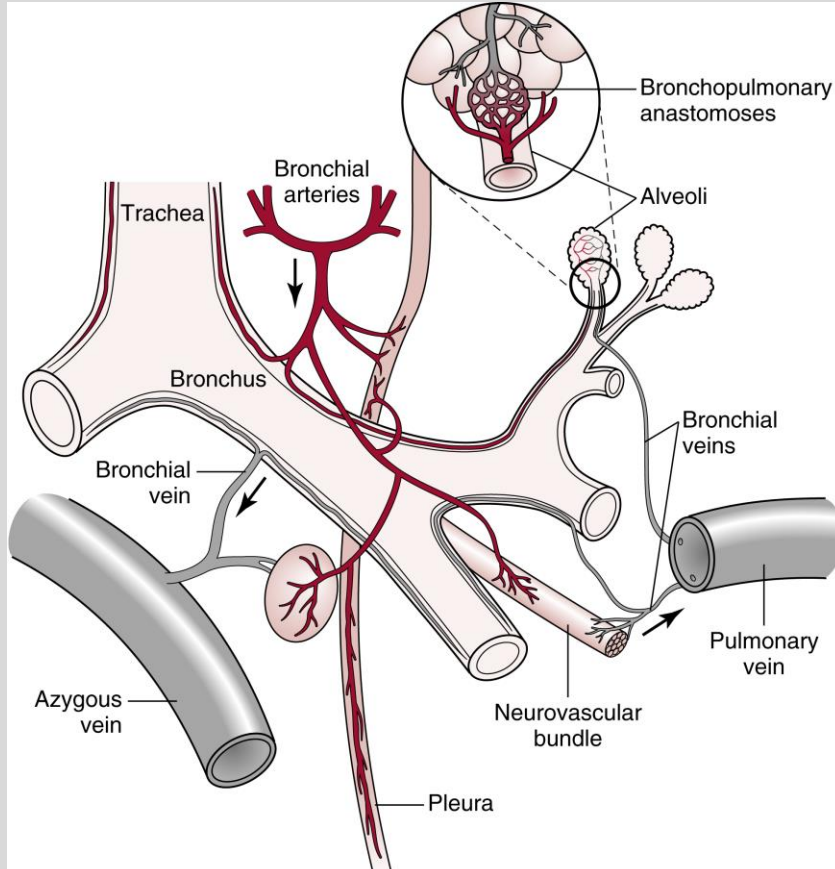
- From Aorta or Intercostal Arteries -



Bronchial Circulation - Backup

Down to the level of terminal bronchioles

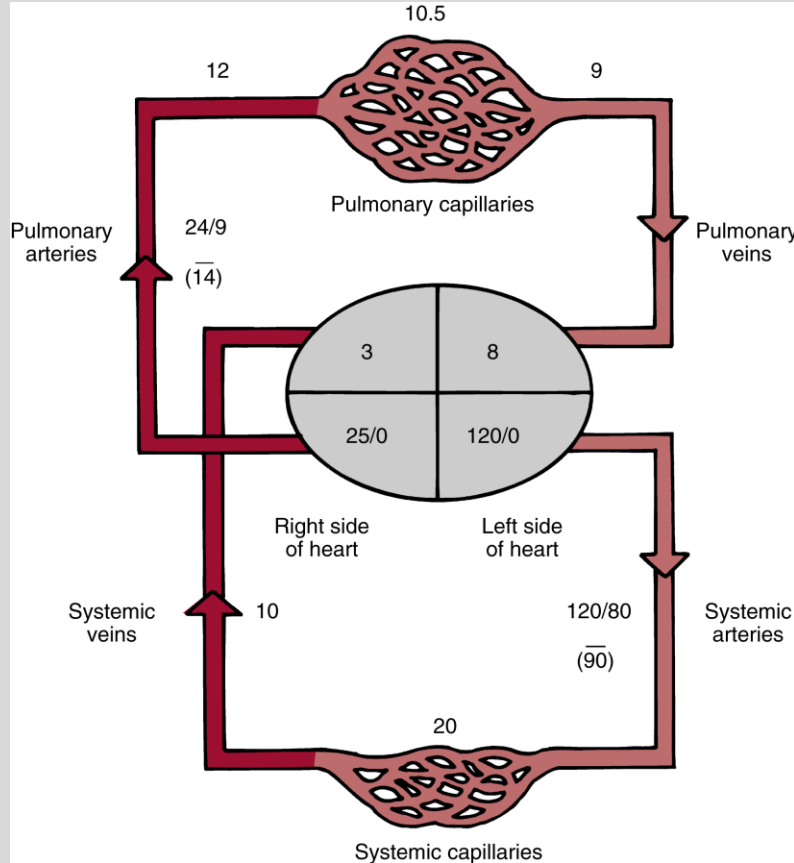
To: trachea and airways, pulmonary blood vessels, visceral pleura, hilar lymph nodes, and vagus nerve and its branches



- Alveoli supplied by **mixed venous circulation**
- **2/3 of blood returned** through **bronchopulmonary anastomoses and veins**
- 1/3 via azygos, hemiazygos, or intercostal veins

Pulmonary & Bronchial Circulation – at Rest Supine

Phasic & Mean Pressures



Dual circulation **reduces effect of embolisms**

Pulmonary Vascular Resistance (PVR)

Calculated via Poiseuille's Law

Influenced through **gravity, alveolar pressure, & arterial to venous pressure gradient**

$$R = \frac{P_1 - P_2}{Q}$$

Resistance = R; Pressure at beginning of tube = P₁;
Pressure at end of tube = P₂; Q = Flow

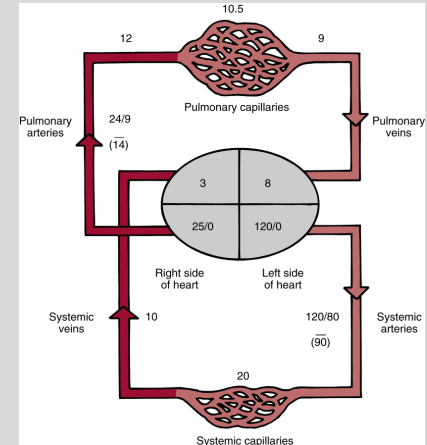
$$PVR = \frac{P_{PA} - P_{LA}}{QT}$$

Pressure in pulmonary artery = P_{PA}; Pressure in left atrium = P_{LA};
QT = Cardiac output = Flow

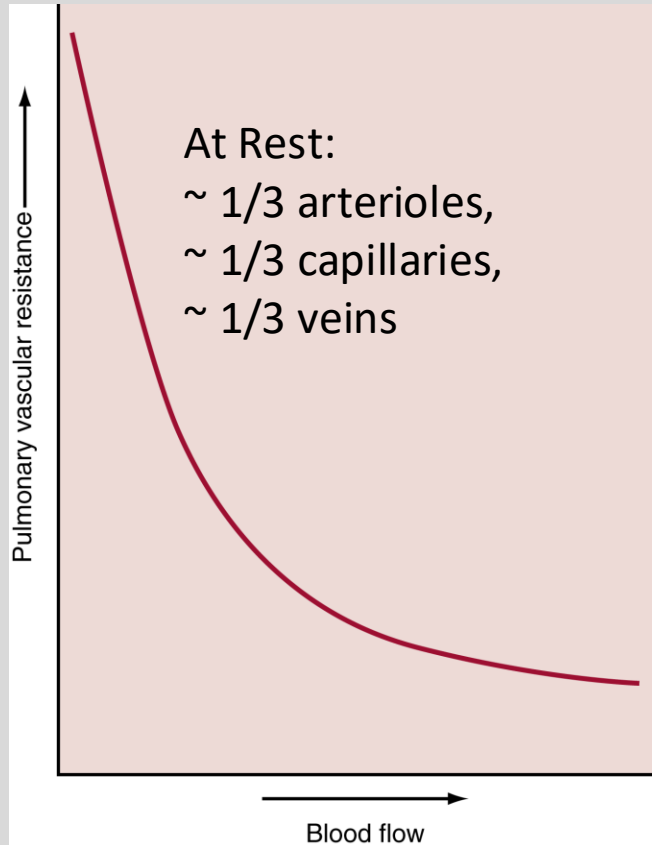
Under normal circumstances this means:

$$PVR = \frac{14\text{mmHg} - 8\text{mmHg}}{6\text{L/min}}$$

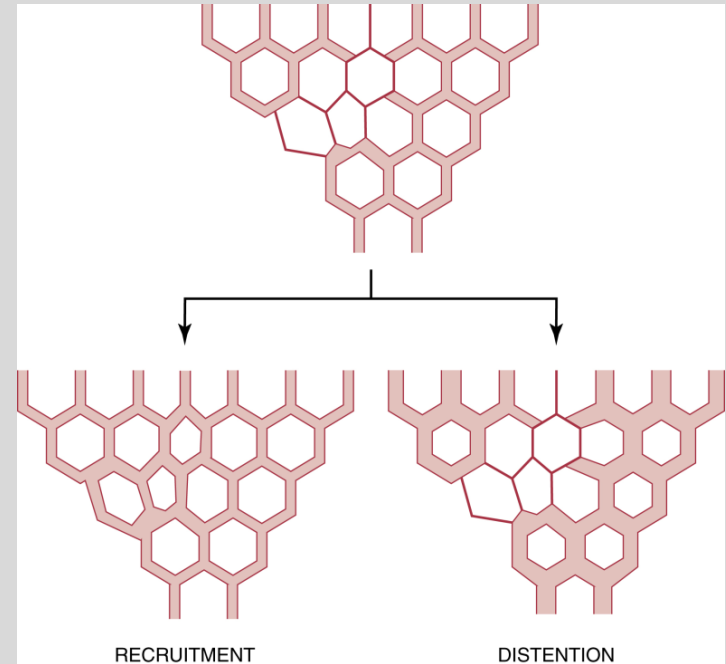
= 1.0mmHg/L/min (**~10 X less than systemic circulation**)



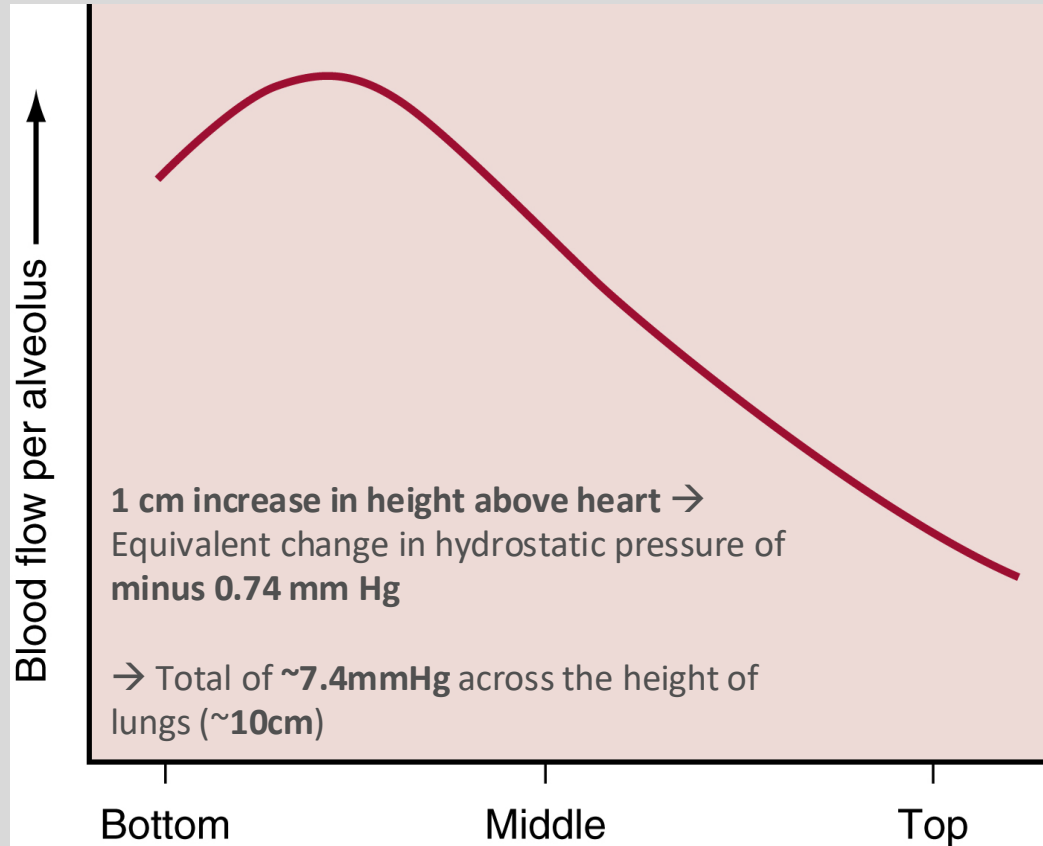
Effect of Pulmonary Blood Flow (Q) on Pulmonary Vascular Resistance (PVR)



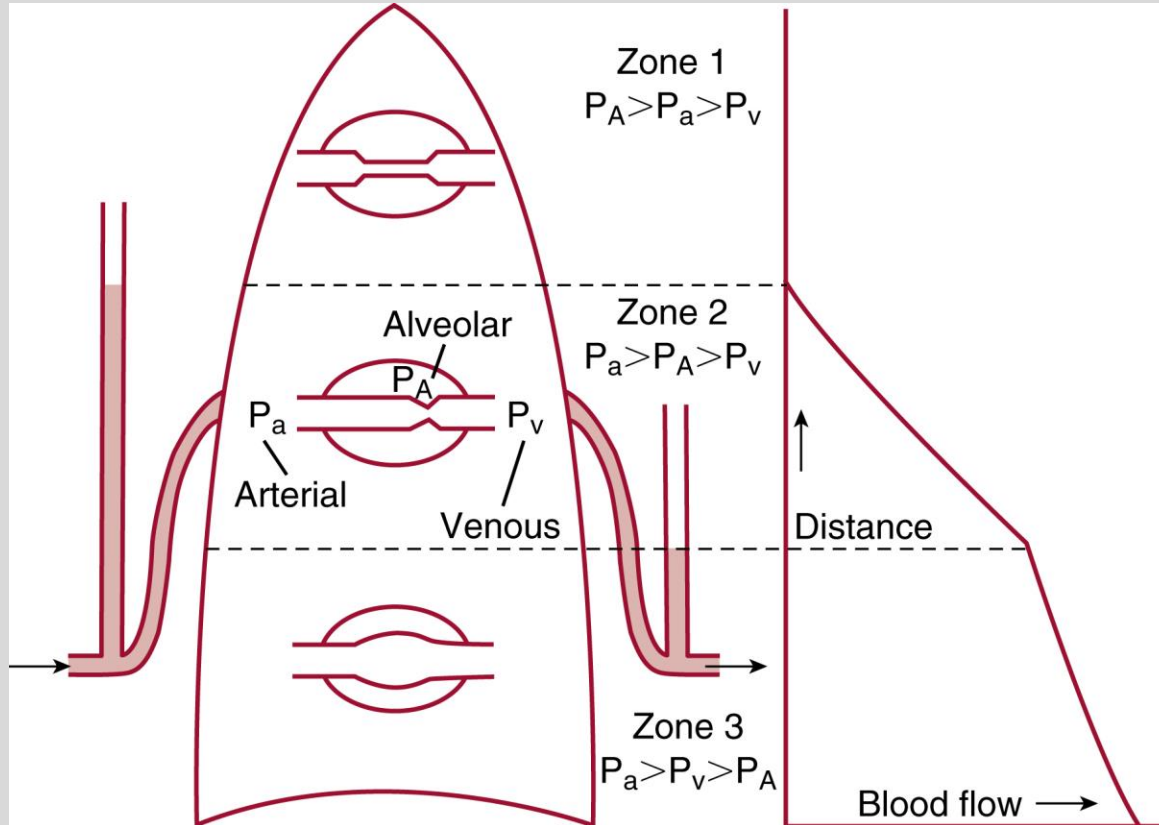
Due to:



The Effect of Gravity on Pulmonary Blood Flow

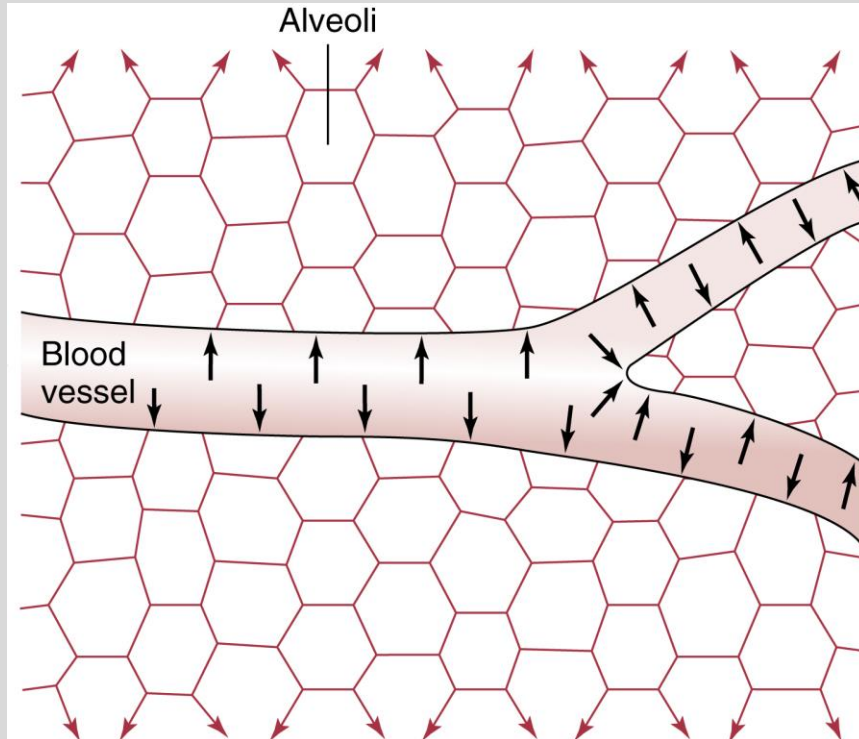


Blood Flow Apex vs. Base – Zones of West



Influence of Pressure on Pulmonary Circulation

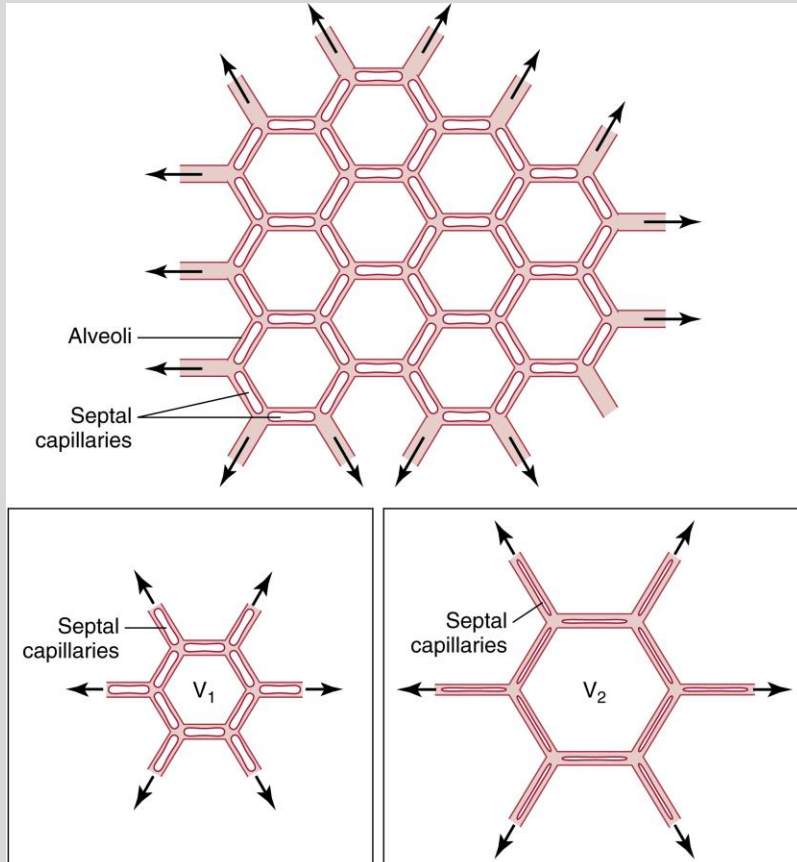
Extraalveolar Vessels (Larger Vessels)



- Alveolar pressure has little effect
- **Pleural & interstitial pressure matter**
- Resistance **high** at low lung volumes and **low** at high lung volumes
- Due to **radial traction**

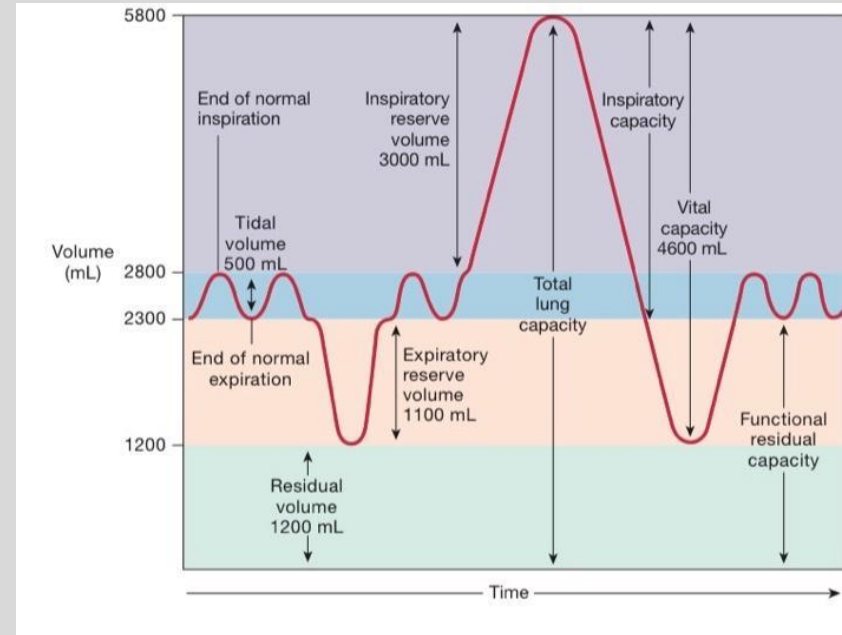
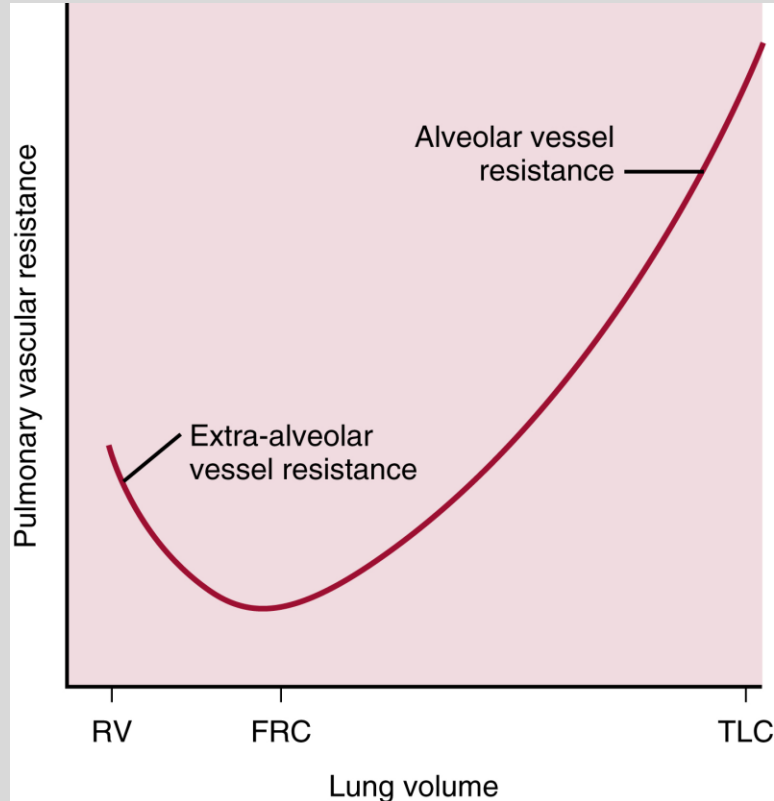
Influence of Pressure on Pulmonary Circulation

Alveolar Vessels (Intraalveolar & Septal Capillaries)



- **Alveolar pressure matters**
- **Pleural & interstitial pressure little effect**
- **Resistance low at low lung volumes and high at high lung volumes**

Total PVR – Combination of Pulmonary & Extrapulmonary Effects



Positive-Pressure Mechanical Ventilation



<https://www.getinge.com/int/insights/articles/intensive-care/diaphragm-monitoring/>

Both

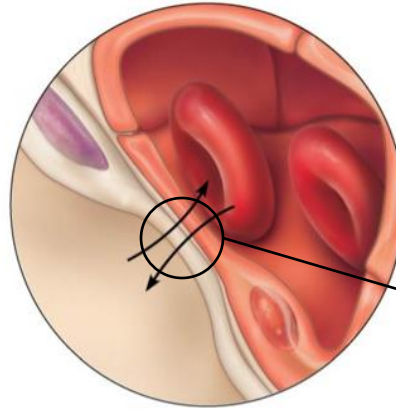
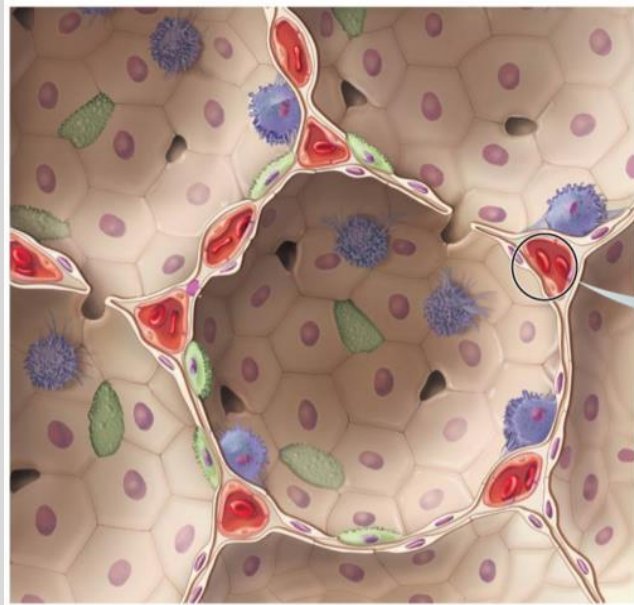
Increase in alveolar pressure &

Decrease in transmural pressure

→ Zone-1 risk

The Alveolar - Capillary Network

Hundreds per Alveolus



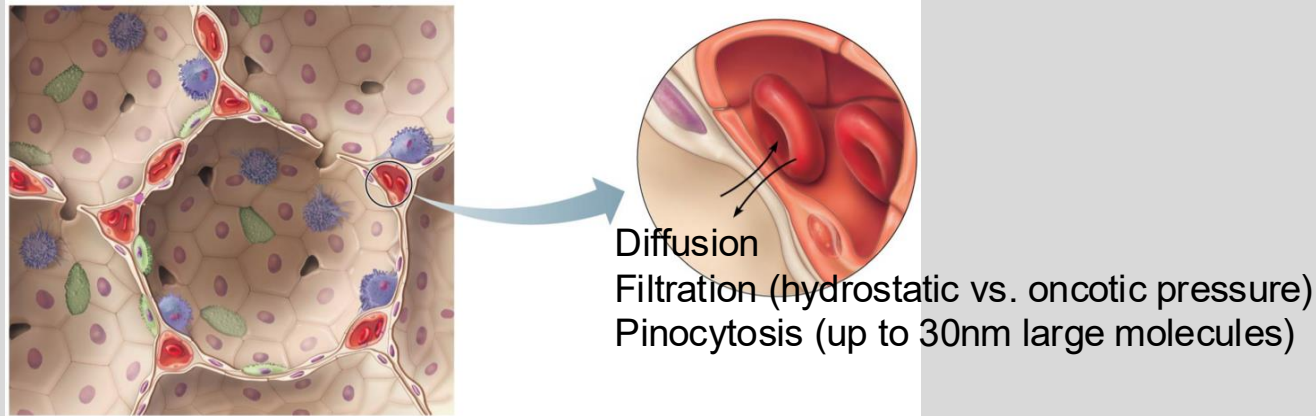
Capillary Endothelium
Basement Membrane (fused)
Alveolar Epithelium

~ 1 to 2 μm thick

Injury of type I epithelial cells $\rightarrow$ type II epithelial cells divide & differentiate into type I

The Alveolar - Capillary Network

Movement of Solutes/Gases

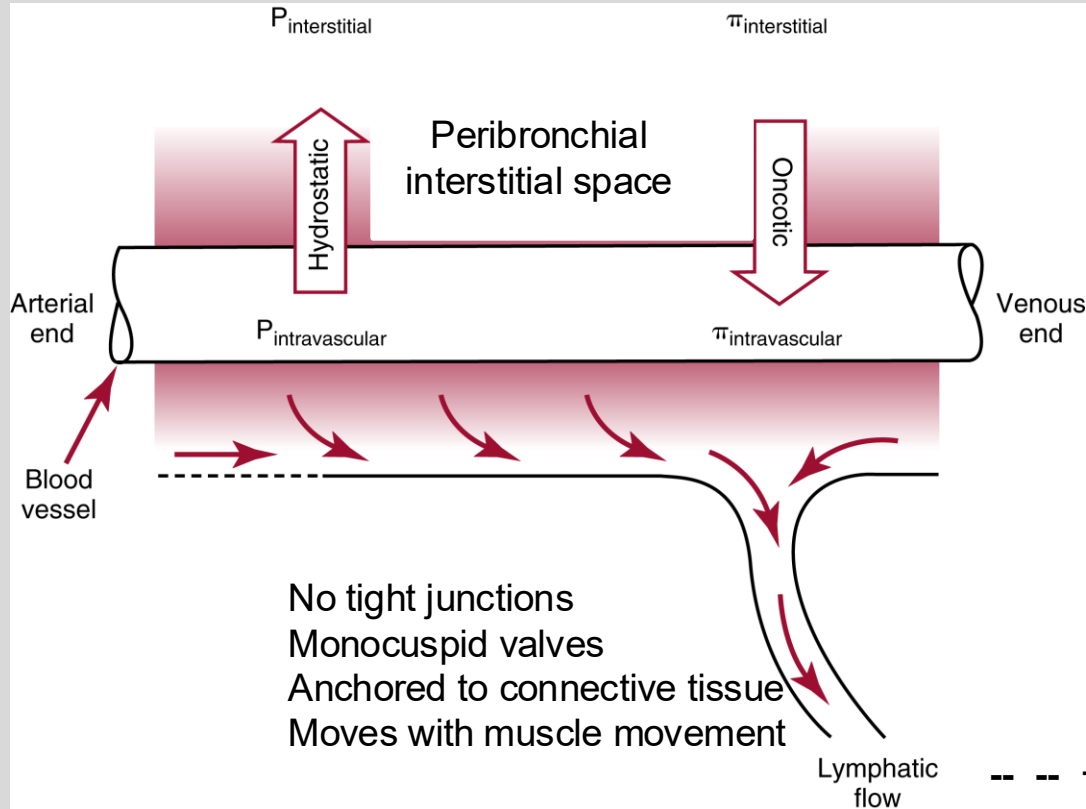


Small molecules: E.g. H_2O , $NaCl$, urea, & glucose – **Flow limited**

Large molecules: **Diffusion limited**

Lipid-soluble molecules: E.g. O_2 , CO_2 – Diffusion through pores or across membranes

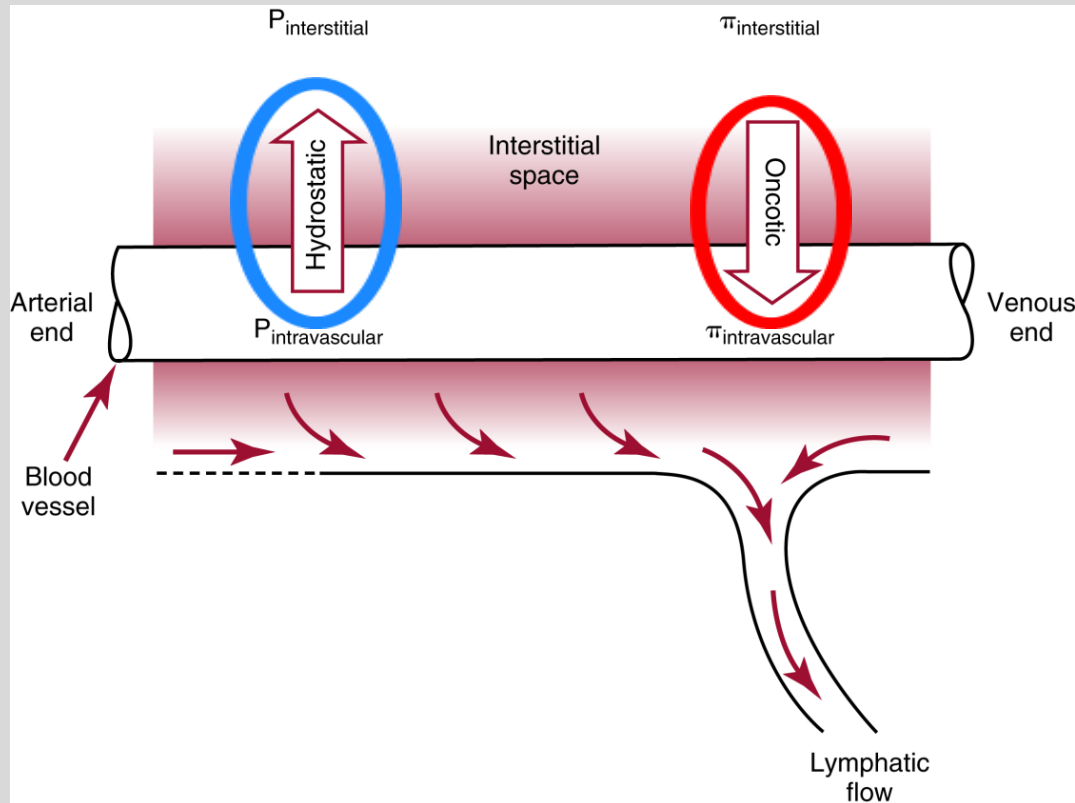
The Pulmonary Lymphatic System



-- -- → Subclavian veins (~30mL/hr)

Pulmonary Edema - Starling Equation

$$\text{Fluid Flux (flow in mL/min)} = K_{fc} [(P_{jv} - P_{is}) - \sigma (\pi_{jv} - \pi_{is})]$$



K_{fc} = capillary filtration coefficient

P_{jv} = intravascular (capillary) **hydrostatic** pressure

P_{is} = interstitial **hydrostatic** pressure

σ = Staveman's reflection coefficient (protein)

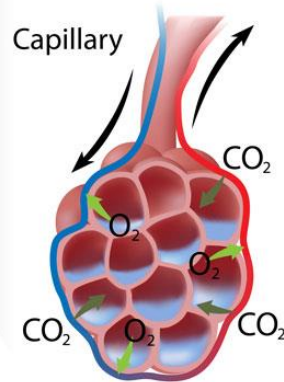
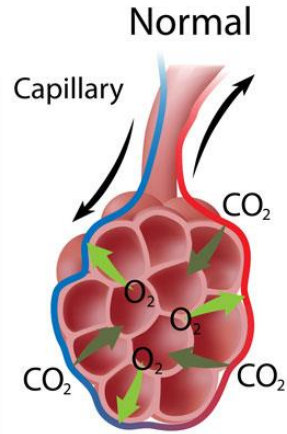
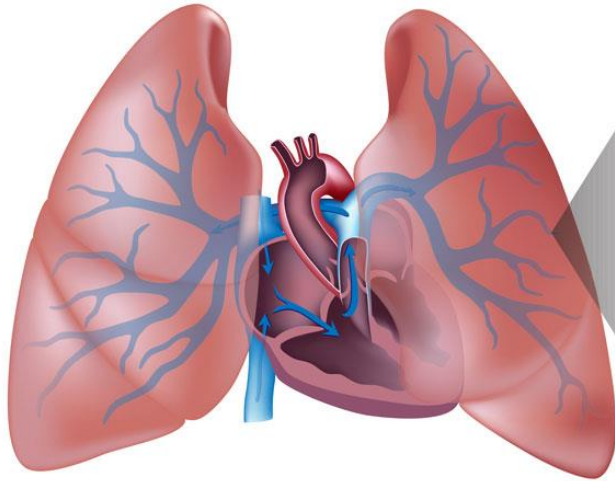
π_{jv} = intravascular **colloid osmotic** pressure

π_{is} = interstitial **colloid osmotic** pressure

σ = lowercase sigma

π = lowercase pi

Pulmonary Edema



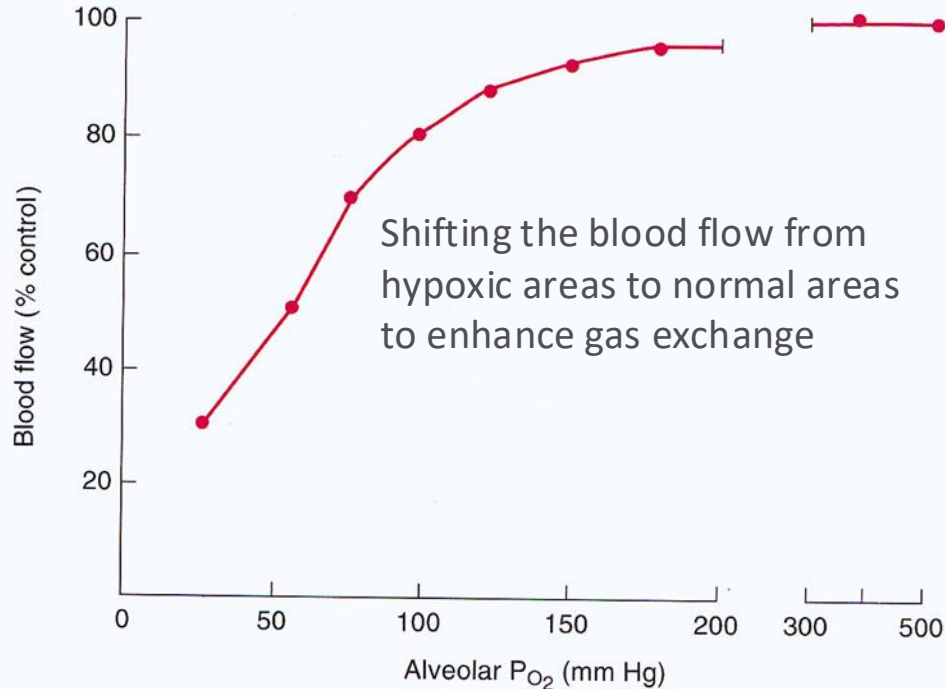
Buildup of fluid
in the air sacs

Increased left atrial pressure

- Through left ventricular infarction/heart failure
- Mitral stenosis
- And other factors...

Active Regulation of Blood Flow

Hypoxic Vasoconstriction



- >20% of vessels before effect on PVR measurable
- High altitude usually more powerful effect

Active Regulation of Blood Flow - Select Pulmonary Vasoconstrictors/Dilators & Airway Constrictors

- **Endothelin** - a **potent vasoconstrictor** released from pulmonary endothelial cells in response to shear stress and hypoxia → Major contributor to pulmonary hypertension
- **Thromboxane A2** - a **potent vasoconstrictor** (arachidonic acid derivative) released from white blood cells, macrophages, and platelets
- **Prostacyclin** (prostaglandin I₂, PGI₂; arachidonic derivative) - a coagulant & **potent vasodilator** released from pulmonary endothelial cells
- **Nitric oxide (NO)** - a **potent vasodilator** released from capillary endothelial cells in response to acetylcholine, ATP, serotonin, bradykinin, histamine, and substance P
- Leukotrienes (inflammatory mediators) - **potent airway constrictors** released by leukocytes

Practice Medical Board-Style Question

A healthy 22-year-old military recruit stands completely upright and motionless at attention during a prolonged morning formation. Due to the effects of gravity on hydrostatic column pressures within the pulmonary vasculature, the distribution of blood flow varies significantly from the apex to the base of her lungs. Which of the following statements best describes the structural and pressure relationships occurring specifically at the apex (Zone 1) of her lungs while she is standing upright?

- A. Alveolar pressure is lower than both pulmonary arterial and venous pressures, maximizing regional perfusion.
- B. Pulmonary arterial pressure is highly elevated by gravity, causing capillary engorgement and maximum gas exchange.
- C. Alveolar pressure can exceed pulmonary arterial pressure, potentially compressing local capillaries and severely limiting blood flow.
- D. Pulmonary venous pressure exceeds alveolar pressure, keeping the vessels fully open regardless of the arterial pressure.
- E. Hydrostatic pressure is at its absolute highest, causing the apical lymphatics to overflow and match the base of the lung in flow volume.

Resources

Updated resources can be found in the Class Preparation Guide (CPG) that goes with this lecture.

Lecture Feedback Form

<https://www.surveymonkey.com/r/NYITCOMLectureFeedback>